

GeoVault AI

Kusbunda OCP

Mine Annual Performance & Operational Report

Financial Year 2023-24

Report ID	GV-KUSMUNDA-OC
Mine ID	P-202324-ANNUAL
Subsidiary	GV002
State	SECL
Coalfield	Chhattisgarh
Mine Type	Korba Coalfields
Dataset Classification	Opencast
Prepared For	SYNTHETIC_DEMO
	GeoVault AI RAG / analytics demonstration

Important: This is a synthetic demonstration report generated from the GeoVault synthetic dataset. It is designed for ingestion, retrieval, analytics, evidence citation and report-generation testing. It is not an official Coal India Limited, CMPDI or subsidiary publication.

1. Executive Summary

During FY 2023-24, Kusmunda OCP recorded actual coal production of 46.25 MT against a target of 49.00 MT, corresponding to achievement of 94.39%. The production variance was -2.75 MT. Year-on-year production growth was 11.18% compared with the preceding financial year. The corresponding offtake was 44.91 MT against an offtake target of 48.27 MT, with offtake achievement of 93.04%. This production/offtake profile provides the primary operational baseline for GeoVault analysis.

The principal recorded operational constraint for FY 2023-24 was High stripping ratio zones increased overburden movement requirements.. The dataset records 13 rainfall-impact days and 173 hours of equipment or operational downtime. Equipment availability was 87.2%, while available manpower was 795. These indicators should be interpreted together rather than treating any single metric as a definitive causal explanation. GeoVault should surface the stated cause alongside the supporting operational indicators so that an analyst can inspect the evidence.

2. Mine Profile

Attribute	Value
Mine	Kusmunda OCP
Mine ID	GV002
Subsidiary	SECL
State	Chhattisgarh
Coalfield	Korba Coalfields
Mine Type	Opencast

Kusmunda OCP is represented in GeoVault as a structured mine entity linked to annual performance records. The profile establishes stable metadata—subsidiary, state, coalfield and mine type—that can be used for filtering, authorization scope and retrieval. Annual metrics are stored separately so that queries such as production trend, target achievement or year-over-year growth can be computed across financial years.

3. Production Performance

Metric	Value
Production Target (MT)	49.00
Actual Production (MT)	46.25
Production Variance (MT)	-2.75
Achievement	94.39%
Previous-Year Production (MT)	41.60
YoY Growth	11.18%
Average Daily Production ('000 t/day)	126.71

During FY 2023-24, Kusmunda OCP recorded actual coal production of 46.25 MT against a target of 49.00 MT, giving achievement of 94.39%. Year-on-year production growth was 11.18%. The variance is retained as an explicit metric for analytical ranking and investigation.

4. Target vs Achievement

Measure	Target	Actual	Achievement
Coal Production (MT)	49.00	46.25	94.39%
Coal Offtake (MT)	48.27	44.91	93.04%

Measure	Target	Actual	Achievement
OB Removal (McuM)	105.84	102.21	See annual record

For FY 2023-24, coal production achievement was 94.39%, while offtake achievement was 93.04%. Production variance was -2.75 MT. GeoVault should preserve these measures independently so that an analyst can distinguish production performance from dispatch performance.

5. Offtake & Dispatch

Actual offtake for Kusmunda OCP was 44.91 MT against an offtake target of 48.27 MT, corresponding to 93.04% achievement. Offtake is stored separately from production because coal produced at the mine and coal dispatched/off-taken are operationally distinct measures. This distinction supports precise SQL and RAG answers.

6. Overburden & Stripping Ratio

Indicator	Target	Actual
Overburden Removal (McuM)	105.84	102.21
Stripping Ratio	2.16	2.21
Rainfall Impact Days	—	13

The FY 2023-24 record shows an actual stripping ratio of 2.21 against a target of 2.16, with 13 rainfall-impact days. Actual overburden removal was 102.21 McuM against 105.84 McuM. These indicators provide operational context but should not be treated as automatic proof of causality.

7. Equipment & Productivity

Indicator	Value
HEMM Availability	93.50%
Equipment Availability	87.20%
Downtime Hours	173
Manpower Available	795
Average Daily Production ('000 t/day)	126.71
Equipment Constraint	N/A

Equipment availability was 87.20% and HEMM availability was 93.50%. The record contains 173 downtime hours and manpower availability of 795. These measures are retained as supporting evidence for production-performance analysis.

8. Operational Constraints

The primary operational cause/constraint recorded for FY 2023-24 is **High stripping ratio zones increased overburden movement requirements..** Supporting indicators include 13 rainfall-impact days and 173 downtime hours. GeoVault should distinguish this recorded cause from analytical inference and cite the underlying mine-year evidence.

9. Geological Observations

OB dump stability and highwall monitoring.

The geological observation is stored as a narrative field for semantic retrieval. In a production deployment, such observations should be linked to the originating geological report, section/page and relevant mine-plan or exploration evidence.

10. Safety Performance

Indicator	Value
Safety Incident Count	7
LTIFR	0.30
Fatalities	0

Safety performance for FY 2023-24 includes 7 recorded safety incidents, LTIFR of 0.30, and 0 fatalities in the synthetic record. These indicators are presented separately from production and environmental performance.

11. Environment & Mine Reclamation

Indicator	Value
Reclamation Area (ha)	32.00
Mine Water Treated (ML)	21.74
Dust Compliance	94.00%

Environmental indicators include 32.00 ha of reclamation, 21.74 ML of mine water treated and 94.00% dust compliance. These measures can support authorized environmental dashboards and cross-year trend analysis.

12. Key Operational Event

No separately recorded event was included for this mine-year.

13. Management Observations & Recommended Follow-up

Recommended analytical follow-up is to compare production achievement (94.39%) and offtake achievement (93.04%) with equipment availability (87.20%), downtime (173 hours), stripping ratio (2.21), and the stated operational constraint. Any management recommendation produced by GeoVault should remain a draft for human review and retain evidence references.

14. Source / Provenance Statement

All numeric and narrative values in this demonstration report originate from the GeoVault synthetic demonstration dataset. The corpus is intended for testing ingestion, chunking, SQL analytics, RAG retrieval, evidence grounding, citation, topic identification and report generation. It must not be represented as official CIL/CMPDI data. The underlying records are marked SYNTHETIC_DEMO.